Jihwan Lee

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Education

University of California, Irvine (GPA: 3.78) **Sep. 2023 – Jan. 2028**

Bachelor of Science in Computer Science *Irvine, California*

- **Relevant Coursework:** Machine Learning, Operating Systems and Compilers, Embedded Systems, Computer Architecture, Digital Logic Design, Graph Algorithms, Data Structures, Software Methodology, Database Management

Experience

University of California, Irvine **March. 2026 - Present**

Undergraduate Researcher *Irvine, California*

- Building a molecular dynamics pipeline in Python using MACE machine-learned potentials to map energy landscapes.
- Integrating force algorithms with generated potential energy surfaces to accelerate sampling of ligand–protein binding.
- Validating simulation accuracy on organic molecules as a testbed toward computational drug-design applications.

University of California, Irvine **Nov. 2025 – Feb. 2026**

ICS 33 - Intermediate Programming Undergraduate Learning Assistant *Irvine, California*

- Supported TAs in labs by assisting students with assignments and projects on advanced Python mechanics.
- Assisted in weekly lectures of 200+ students, providing individualized support for the students.
- Facilitated learning online by monitoring and answering students' questions on class discussion boards.

Projects

GameSetMatch | *TypeScript, React, Postgres, Next.js, Python, Scikit-Learn*
https://gamesetmatch.vercel.app/

- Built a tennis match history web app with TypeScript using Postgres, with a Next.js and React frontend.
- Created a Python ETL pipeline ingesting 80K+ matches from a public archive into Postgres database.
- Trained a Scikit-Learn performance-rating model and a surface-aware Elo match predictor validated at 70% accuracy.

Tennis Stroke Comparison | *Python, MediaPipe, OpenCV, JavaScript, HTML5 Canvas*
https://jihwantime.github.io/tennis-stroke-comparison/

- Built a Python CLI that extracts 33-keypoint pose skeletons from tennis video using MediaPipe Pose Landmarker.
- Engineered frame-accurate skeleton overlays on raw video using HTML5 Canvas and requestVideoFrameCallback.

Diabetes Patient Readmission Predictor | *Python, Pandas, Scikit-Learn, Matplotlib*
https://colab.research.google.com/drive/1gzmmgKpRib6UthHhqpDiMU6IR8Z8EkX2?usp=sharing

- Trained and evaluated an array of Scikit-Learn classification algorithms to predict patient readmission.
- Implemented hyperparameter tuning and adjusted decision thresholds to address severe class imbalances within the data.
- Tuned model correctly identified 80% of at-risk patients that were overlooked by baseline model.

SQL Airport Database Project | *Python, SQL*
http://learning-to-fly.onrender.com/

- Developed an event-based Python backend engine to bridge tkinter GUI with an SQLite database.
- Implemented secure SQL queries to perform CRUD operations across multiple relational database tables.

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript, HTML/CSS, Verilog, Cuda, SQL
Frameworks: React, Tailwind, Hono, Scikit-Learn, NumPy, Matploblib, Pandas, OpenCV, MediaPipe
Interests: Tennis, Bouldering, Music, Camping/Hiking, League of Legends

Leadership / Extracurricular

Tennis Club **Spring 2025 – Present**

Vice President *University of California, Irvine*

- Part of the Executive Board. Main job is to supervise and provide guidance for the Competitive Team.
- Address concerns brought up between competitive board members or competitive team members, and executive board.
- Work with fundraiser chair to raise 75% of our annual Sport Club allocation funds.